TRIBHUVAN UNIVERSITY

INSTITUTE OF SCIENCE AND TECHNOLOGY

A Project Proposal On

RESUME-DRIVEN ADAPTIVE INTERVIEW PRACTICE SYSTEM

Submitted To:

DEPARTMENT OF CSIT

NATIONAL COLLEGE OF COMPUTER STUDIES(NCCS)

PAKNAJOL, KATHMANDU

In partial fulfilment of the requirements for the degree of Bachelors of Science in Computer Science and Information Technology (B.Sc. CSIT)

Submitted By:

Nabin Yonjan (79010733)

February, 2026

TABLE OF CONTENT

LIST OF FIGURES

1. INTRODUCTION

Interview preparation is an important step for students and job seekers aiming to enter technical fields. However, most existing interview practice platforms follow a generic approach by presenting the same set of questions to all users, regardless of their skill level or background. Such systems limit effective self-assessment and do not provide candidates with meaningful insights into their actual technical proficiency.

With recent advancements in artificial intelligence (AI) and natural language processing (NLP), researchers have explored intelligent systems that support resume analysis and personalized interview preparation. Studies show that AI-based platforms can extract skills from resumes and assist candidates by providing structured interview practice and feedback, improving preparation efficiency and learning outcomes [1] [2].

Despite these developments, many existing systems do not dynamically adjust question difficulty based on user performance or provide clear skill-wise performance evaluation. This project proposes a web-based adaptive interview practice and skill assessment system that analyzes a candidate’s resume, conducts performance-driven interview practice, and generates detailed feedback to support effective interview preparation and self-evaluation.

2. PROBLEM STATEMENT

Interview preparation is essential for students and job seekers, but most existing platforms offer generic, non-adaptive questions that do not match a user’s skills or experience. Candidates often waste time on irrelevant questions and receive limited feedback on their performance. Current systems also fail to adjust question difficulty based on real-time responses, making it difficult to identify strengths and weaknesses.

This project proposes a **resume-driven, adaptive interview practice and skill assessment system** that selects questions relevant to a candidate’s skills, adjusts difficulty dynamically, and provides **structured, skill-wise feedback** to improve preparation and self-assessment.

3. OBJECTIVES

The main objectives of the project are:

1. **Resume Analysis:** To extract key technical skills from a candidate's resume for personalized assessment.
2. **Adaptive Question Selection:** To select interview questions based on identified skills and dynamically adjust difficulty according to user performance.
3. **Skill-Wise Feedback:** To provide detailed feedback on strengths, weaknesses, and performance across different difficulty levels.
4. **Performance Visualization:** To display skill-wise and difficulty-wise performance using charts for easy interpretation.
5. **Effective Interview Practice:** To help candidates improve their readiness and self-assessment before real interviews.

4. SCOPE AND LIMITATION

4.1. Scope

- The system targets students and job seekers preparing for technical interviews.
- It allows users to upload resumes, analyze skills, and receive **personalized interview questions**.
- Questions are **categorized by skill and difficulty** and dynamically adjusted based on user performance.
- The system provides **detailed feedback** and **performance visualization**, helping users identify strengths and weaknesses.
- The project is **web-based**, making it accessible from anywhere without installation.

4.2. Limitations

- The system relies on a **predefined question bank**, so it may not cover every possible topic or scenario.
- Resume parsing uses **basic keyword-based NLP**, which may not capture complex resume structures or implicit skills.
- Adaptive difficulty logic is **rule-based**, not powered by deep machine learning, which may limit accuracy in some cases.
- The system is designed for **practice purposes only** and does not guarantee success in real interviews.

5. METHODOLOGY

The methodology outlines the systematic approach for designing and developing a resume-driven adaptive interview practice and skill assessment system. It includes requirement identification, feasibility study, proposed system design, and scheduling for implementation.

5.1. Requirement Identification

5.1.1 Study of Existing Systems / Literature Review

In 2022, the *AI-Powered Resume Analyzer and Interview Preparation System* [1] proposed a framework for extracting skills from resumes and generating relevant interview questions. The system uses NLP techniques to analyze resumes and provides structured feedback on performance. However, it relies on **static question sets** and does not adjust question difficulty dynamically based on user performance.

In 2023, Harsh Koshti et al. [2] developed an AI-based interview preparation platform that offers personalized mock interviews and feedback using candidate skill profiles. While the system emphasizes structured practice, it lacks **skill-wise performance evaluation** and does not dynamically adapt to individual responses.

The **AI-Based Resume Analyzer and Interview Simulator** [3] uses NLP for resume parsing, machine learning for role mapping, and adaptive question generation to provide personalized interview simulations. Its workflow includes resume upload, role selection, AI-driven interview, real-time feedback, and improvement planning, helping candidates identify skill gaps and improve readiness efficiently.

These studies highlight a gap in current systems, demonstrating the need for a platform that **combines resume analysis, adaptive question selection, and detailed skill-wise feedback**, forming the foundation for the proposed project.

5.1.2 Requirement Collection

The requirements for the proposed system were collected through **analysis of existing solutions and examination of user needs**. The key requirements identified include:

1. **Resume Analysis:** Ability to parse resumes and extract relevant technical skills.
2. **Adaptive Question Selection:** Generate interview questions based on skills and adjust difficulty according to user performance.
3. **Feedback and Performance Evaluation:** Provide detailed skill-wise feedback and performance metrics.
4. **Web-Based Access:** The system should be accessible through browsers without installation.
5. **User-Friendly Interface:** Simple navigation and clear presentation of questions and feedback.

5.2. Feasibility Study

The feasibility study evaluates whether the proposed **adaptive interview preparation system** can be successfully designed, implemented, and used. Feasibility is analyzed from **technical, operational, and economical** perspectives.

5.2.1 Technical Feasibility

The proposed system is technically feasible because it uses **well-established, accessible technologies** and custom algorithms:

- **Web-based platform:** Built with HTML, CSS, JavaScript, and Python (Flask/Django) for backend.
- **Database:** MySQL stores questions, user performance, and resume data.
- **Custom resume parser:** Implements text extraction, tokenization, and keyword matching from scratch.
- **Adaptive algorithm:** Adjusts question difficulty dynamically based on user performance.
- **Optional ML classification:** Can classify skill proficiency (strong/medium/weak) to meet AI/ML requirements.
- **Visualization:** Charts display skill-wise feedback clearly.

The **core intelligence** is in the adaptive algorithm and feedback logic, while supporting tools handle data storage and preprocessing.

5.2.2. Operational Feasibility

The system is operationally feasible because:

- It is **web-based**, accessible from any device with a browser.
- Users can **upload resumes, practice interview questions, and receive feedback** without technical expertise.
- The adaptive question mechanism allows users to **practice efficiently and assess their skill strengths and weaknesses**, increasing usability.
- Skill-wise feedback charts provide **clear, actionable insights** for improvement.

5.2.3. Economical Feasibility

The project is economically feasible because:

- Development uses **open-source tools and libraries**, minimizing software costs.
- Deployment can be done on a **basic web server or low-cost cloud service**.
- Maintenance costs are minimal since the system is designed primarily for **self-practice and learning**, with no heavy computational requirements.

5.2.4 Schedule (Gantt Chart)

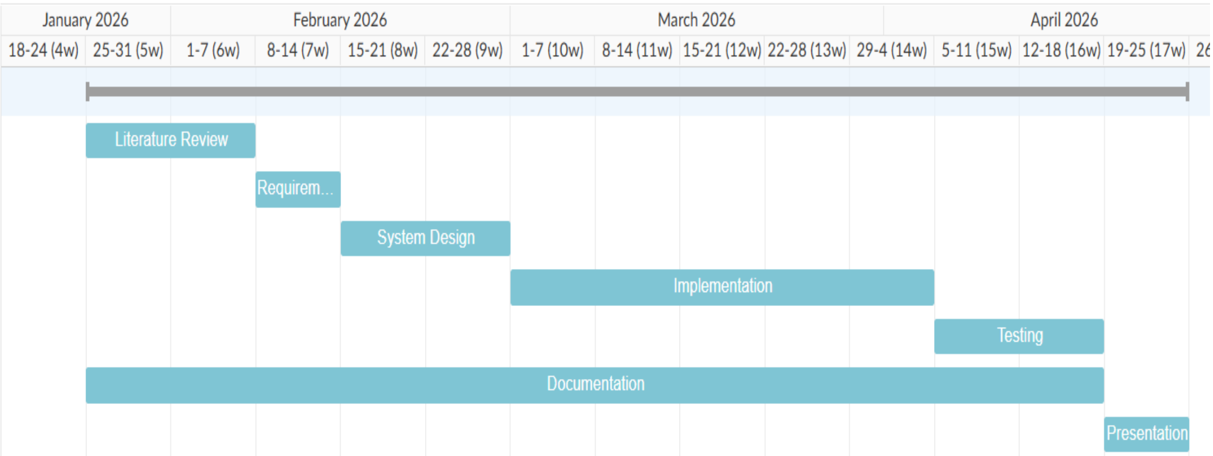

Figure 1:Gantt Chart

5.3. High-Level Design of Proposed System

The proposed system is a **web-based adaptive interview practice platform** that allows users to **upload resumes, attempt skill-based interview questions, and receive personalized feedback**. The system's intelligence is based on an **adaptive algorithm** that selects questions dynamically based on user performance and optional inferred confidence. NLP or preprocessing libraries are used only to **parse resumes**, while the core reasoning is **algorithmic AI**.

5.3.1 System Modules

❖ **User Interface Module**

- Web-based interface for login, resume upload, practice questions, and viewing feedback.
- Clean, user-friendly navigation ensures easy access for users of all levels.

❖ **Resume Parsing Module**

- Extracts text from uploaded resumes (PDF/TXT).
- Cleans and tokenizes text, matches tokens with **predefined skill keywords**, and stores detected skills in the database.
- This allows **questions to be tailored to the candidate's skills**.

❖ **Question Bank Module**

- Stores **pre-curated questions** categorized by skill and difficulty (easy, medium, hard).
- Can be expanded with more questions without changing the system logic.

❖ **Adaptive Question Selection Module**

- Selects questions dynamically based on **Performance Score**, considering:
  - Correctness of answer
  - Response time
  - Optional inferred confidence (user-friendly, inferred from speed)
- Adjusts next question difficulty accordingly.

❖ Feedback & Skill Classification Module

- Generates skill-wise performance feedback after the session.
- Displays visual charts for strengths and weaknesses.
- Optionally classifies skill proficiency as Strong, Medium, or Weak.

❖ Database Module

- MySQL stores user profiles, resumes, questions, and performance results.

5.3.2 System Workflow

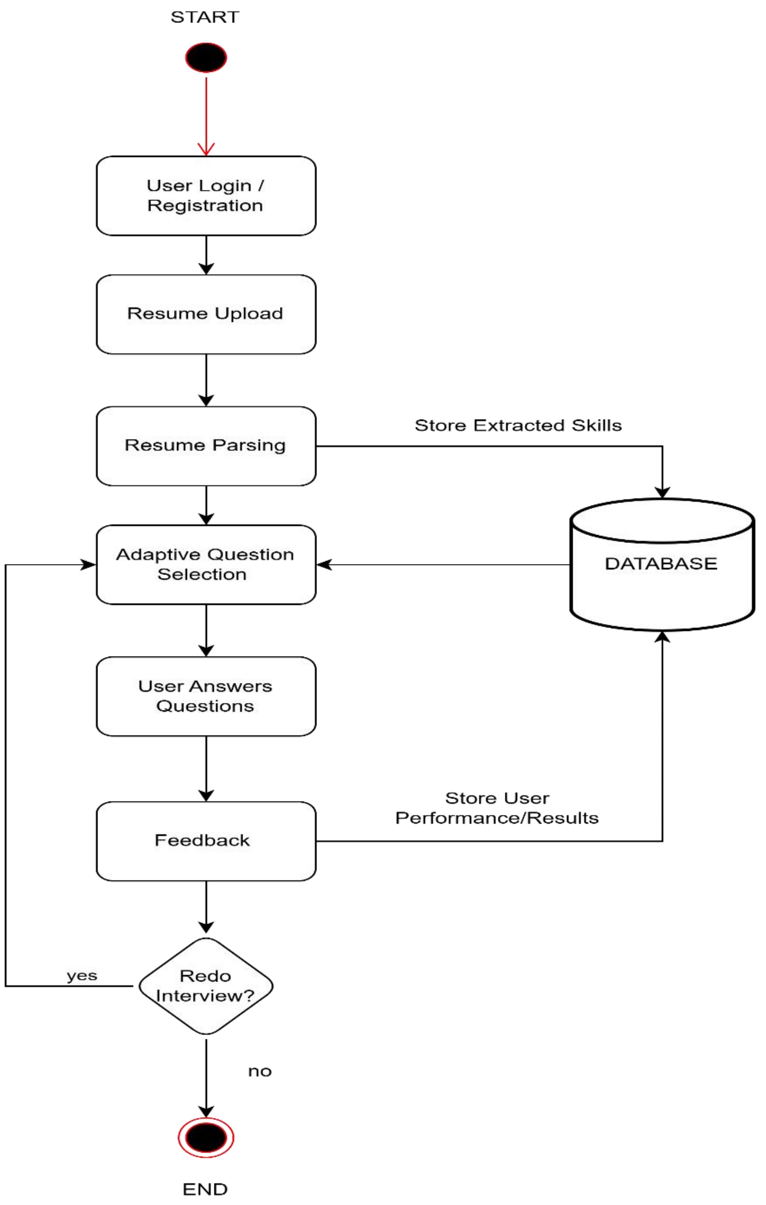

Figure 2:Flow Chart

5.3.3 Description of Algorithms

A. Resume Parsing Algorithm (Custom + NLP-enhanced)

1. Read resume text (PDF/TXT) using lightweight libraries.
2. Clean text (lowercase, remove punctuation, special characters).
3. Tokenize words and phrases using custom logic.
4. Optional NLP enhancement:
    - Named Entity Recognition (NER) for skills, degrees, job titles, tools.
    - Context-based skill detection (e.g., “developed ML model” → Machine Learning).
5. Compare tokens/entities with predefined skill keywords and map to question categories.
6. Store matched skills in the database.

B. Adaptive Question Selection Algorithm

For each skill:

1. Start with an easy question.
2. After user answers, calculate **PerformanceScore**:

$$PerformanceScore = \alpha * Correctness + \beta * ResponseTime + \gamma * Confidence$$

3. Decide next question difficulty:

$$NextDifficulty = CurrentDifficulty + f(PerformanceScore)$$

    - High PerformanceScore → harder question
    - Low PerformanceScore → easier or repeat question

4. Continue until all questions are completed or session ends.

C. Skill Classification & Feedback Algorithm

For each skill:

1. Calculate percentage of correct answers.
2. Assign proficiency:

    - ≥80% → Strong
    - 50–79% → Medium
    - <50% → Weak
3. Display results using visual charts or tables.

Feedback module is mandatory for the adaptive system, providing insights into strengths and weaknesses.

6. EXPECTED OUTCOMES

The proposed system aims to provide a comprehensive, adaptive interview practice platform. After implementation, the expected outcomes are:

1. **Adaptive Questioning**
   - Questions dynamically adjust to the user's skill level using PerformanceScore, ensuring an optimal learning curve.
   - Users are challenged appropriately without being overwhelmed.
2. **Personalized Skill Assessment**
   - Resume parsing (custom + NLP-enhanced) identifies candidate skills accurately.
   - Questions are tailored to the candidate's specific technical and domain skills.
3. **Feedback & Performance Insights**
   - After each session, users receive skill-wise feedback, highlighting strengths and weaknesses.
   - Skill proficiency is classified as Strong, Medium, or Weak, and performance charts visualize results for easy understanding.
4. **Data-Driven Improvement**
   - Users can track progress over multiple sessions.
   - Provides suggestions for areas to focus on for skill improvement.
5. **Scalable and Extensible System**
   - New questions and skills can be added to the **Question Bank** without changing system logic.
   - The system can be extended to support **additional roles, programming languages, or skill areas** in the future.
6. **Real-World Impact**
   - Helps candidates **prepare efficiently for interviews**.
   - Bridges the gap between learned skills and practical assessment.
   - Provides organizations or educational institutions with a **tool for skill assessment and training**.

7. REFERENCES

[1] P. K. R. P. K. B. P. A. K. V. V. V. Ramesh B E, "ijraset," [Online]. Available: https://www.ijraset.com/research-paper/ai-powered-resume-analyzer-and-interview-preparation-system. [Accessed 1 2 2026].

[2] P. G. R. P. P. K. S. T. Harsh Koshti, "journaljerr," [Online]. Available: https://journaljerr.com/index.php/JERR/article/view/1489/3223. [Accessed 1 February 2026].

[3] M. M. R. M. J. P. P. N. S. R. L. Mayank Chauhan, "zenodo," [Online]. Available: https://zenodo.org/records/17875851. [Accessed 1 February 2026].